the Device Mapper (See Figure 3). The Enterprise Virtual Management System also has dependency on Device Mapper for the basic function of mapping a block device. On Linux, *dmsetup*, which is a command line wrapper to access Device Mapper functionality, is available.

Network-based virtualisation: Technologies such as Storage Area Networks allow hard disks and tapes to be available to multiple servers. Almost all operating systems for the enterprises have the ability to connect to a SAN either over IP or fibre channel. This also simplifies the management of storage hardware from different vendors.

SCSI has been a standard protocol for computing devices to communicate with the peripherals. The same communication has been extended for communication over the well-established Internet protocol, TCP/IP. This is iSCSI, which is the low cost alternative to the fibre channel. Due to the inherent computational overhead of TCP/IP, the iSCSI communication can cause a high load on the CPU. To overcome this limitation, some vendors came up with TCP Offload engines. One of the open source implementations of iSCSI network protocol is the open-iSCSI project, which is a high-performance implementation of RFC3720.

Storage device virtualisation: This is where the latest technology updates are happening. Storage vendors are bringing innovative solutions to market at a rapid pace. Virtualisation on storage devices can be at the block level or file level. The block level virtualisation can be found in intelligent disk subsystems. The servers access the storage through the I/O channels by means of LUN masking and RAID. The common protocols that run between the server and storage are SCSI, FCoE and iSCSI.

Virtualisation at the file level is available via NAS servers, which take the responsibility of file system management. Some of the free or open source NAS products are Openfiler, FreeNAS and CryptoNAS. Openfiler has been covered in a series that started in the August 2011 issue of *LINUX For You*. FreeNAS has extensive features for networking, services, drive management and monitoring. It also has advanced features such as snapshots and thin provisioning. CryptoNAS has been developed with disk encryption as the focus. It comes in two flavours – CryptoNAS-Server and CryptoNAS-CD. CryptoNAS was earlier called CryptoBox.

There are several advantages of storage devices. Servers are freed up from the CPU-intensive tasks of virtualisation operations and applications perform better as RAID operations are done by storage controllers. The administration of storage devices can be done in isolation and nearer to the physical devices. Heterogeneous (multi-vendor) storage devices can be deployed to get the best of each vendor technology. While this is an advantage, sometimes the proprietary features of products from some vendors can make setting up a solution an uphill task.

Storage virtualisation in enterprise environments: In a data centre, when you migrate from physical servers to virtual servers, you could gain on CPU utilisation. The physical servers that were used in the range of 20 to 60 per cent are now placed as VMs on a single physical machine and could maximise server utilisation. Server utilisation also means that the read-write access density increases on the storage. Addressing this challenge is what decides the success of storage virtualisation.

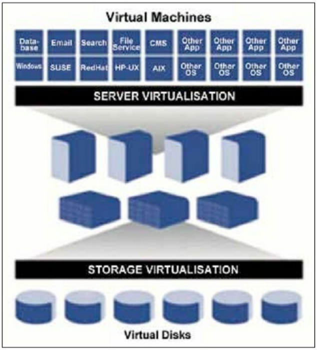

Figure 1: The server virtualisation and storage virtualisation layers

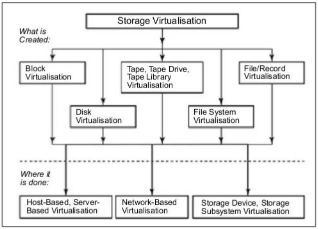

Figure 2: Three locations of storage virtualisation (courtesy: SNIA – Virtualisation taxonomy)

In enterprise environments, there is the constant challenge of ensuring the performance of business critical applications and the increasing demand for vast storage capacity, while keeping the overall costs as low as possible.

Hierarchical storage management: Hierarchical storage management (also known as data lifecycle management) is a concept in which the most recent data is stored in storage subsystems that are quick to access, and stored on types of media that offer the fastest access. As the